Seismic Slope Stability Analysis in GIS: A Case Study for the Ok Tedi Mining Area, Papua New Guinea

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BACKGROUND

What is Seismic Slope Stability Analysis in GIS?

Seismic Slope Stability Analysis in GIS refers to the process of evaluating the stability of slopes (such as hillsides, embankments, or natural terrain) under seismic (earthquake) loading using Geographic Information Systems (GIS) tools and spatial data. The analysis integrates seismic hazard data with terrain characteristics to assess the likelihood of slope failure or landslides during seismic events.

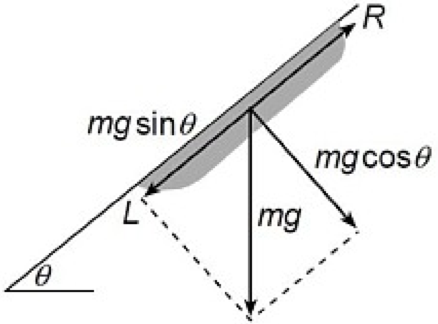
This can help in assessing and managing the seismic-induced landslide risks over large areas more effectively. This process involves creating detailed maps that show how local geological, geomorphological, and structural conditions influence the ground motion during an earthquake, which helps identify areas of higher seismic hazard.

Case Study: Ok Tedi Mining Area

This sample project was undertaken as a major project in the Master of Engineering Studies program at the University of Auckland. Geological, topographical, geotechnical and seismic data from the Ok Tedi area were used to generate maps identifying the most likely areas prone to landslide.

METHOD

Wilson et al. (1979)



The Wilson et al. method assumes that in a thin layer, sliding occurs due to inertial loading.

$$a_c = g \cdot \left[\frac{c}{\gamma h} + \left(\cos\theta \tan\phi - \sin\theta \right) \right]$$

Where: a_c *= critical acceleration inducing sliding, g = acceleration due to gravity, c = cohesion of soil,* ϕ *= internal friction angle of the layer,* γ *= unit weight of soil,* θ *= slope angle, and* h *= thickness of the sliding layer.*

WORKFLOW

Topographic Contour → Slope Angle (θ) Map

Geologic Map → Input Geotechnical Properties (γ, c, ϕ)

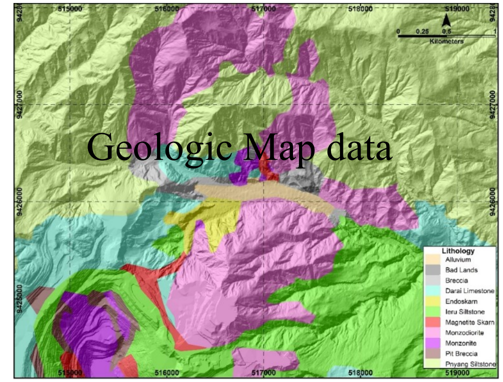
Thickness (h)

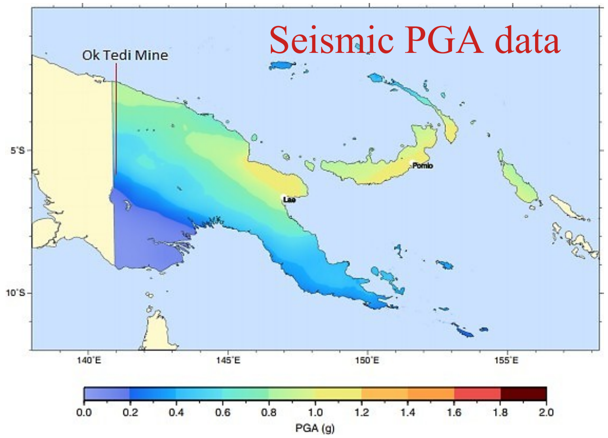
→ Compute Critical Acceleration (a_c)

$$a_c = g \cdot \left[\frac{c}{\gamma h} + \left(\cos\theta \tan\phi - \sin\theta \right) \right]$$

→ $$FoS = \frac{a_c}{PGA}$$

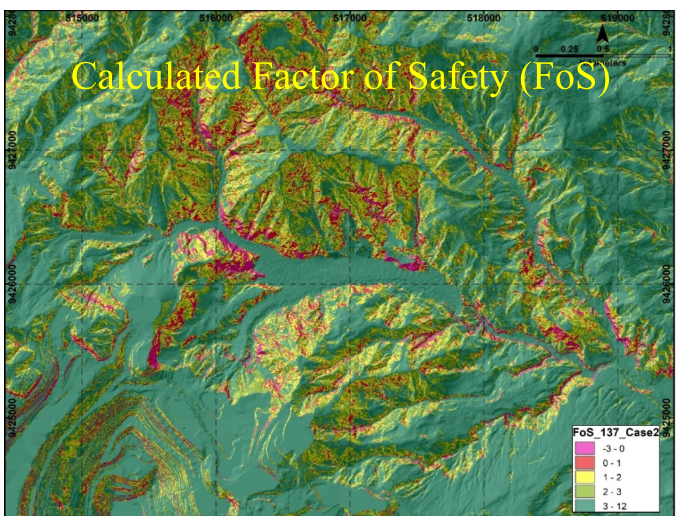
→ Risk Ranking





Geotechnical Data

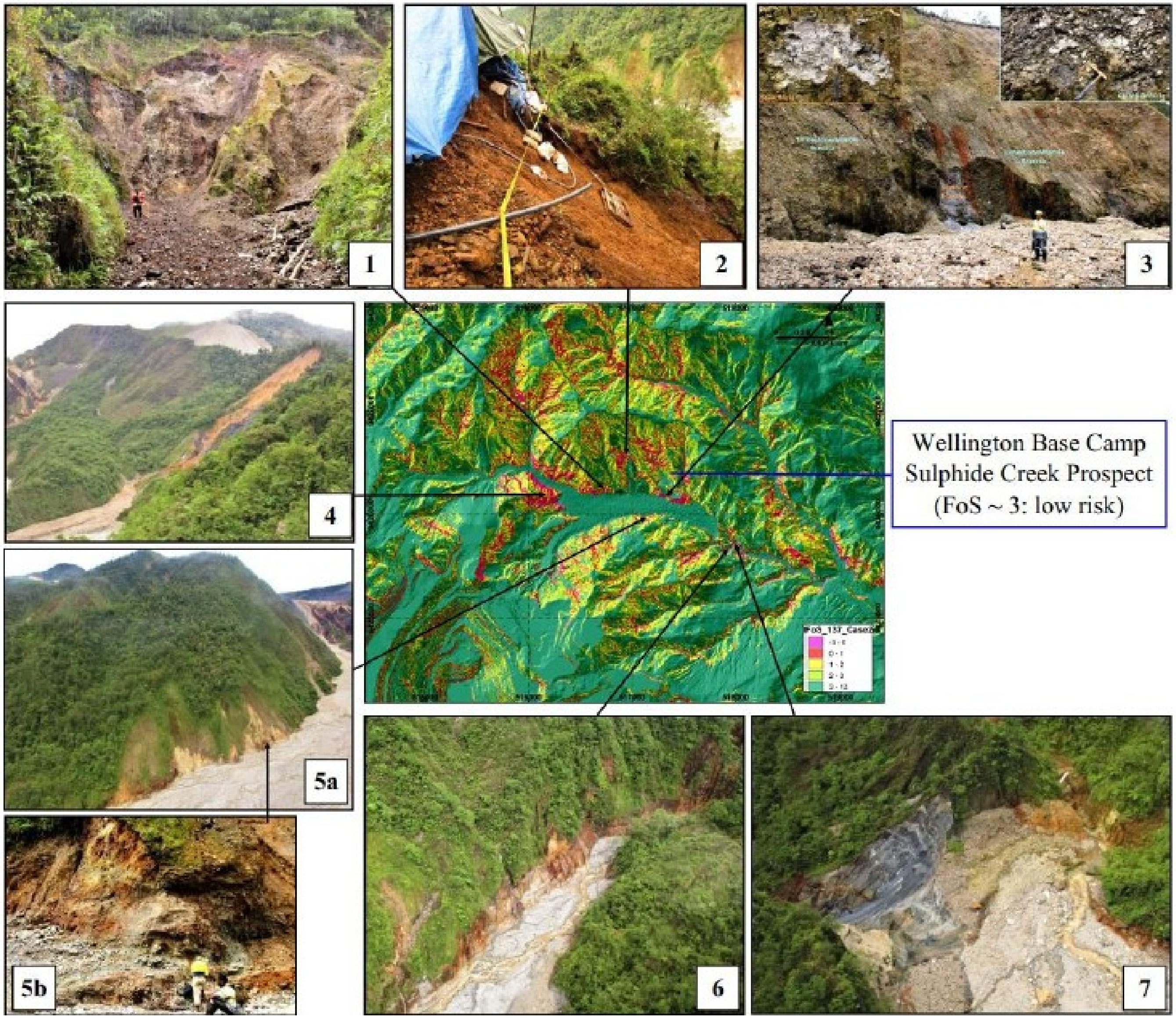
Lithology	Unit Weight	Friction	Cohesion	Thickness (m)	
	(kN/m³)	(Deg)	(kPa)	Case 1	Case 2
Pnyang Siltstone	26.6	32	90	2	5
Ieru Siltstone	26.2	33	90	2	5
Monzonite	25.5	32	70	2	5
Darai Limestone	27.5	31.5	80	2	5
Monzodiorite	25.5	32	70	2	5
Pit Breccia	21	31	60	2	5
Bad Lands	23	12	30	2	2
Magnetite Skarn	44.5	45	200	2	5
Alluvium	24	18	40	2	2
Breccia	21	27	60	2	5
Endoskarn	32.5	40	100	2	5



Risk Ranking based on FoS

FoS	Risk Rank
< 1	Predicted Slope Failure
1 – 2	High Risk
2 – 3	Moderate Risk
> 3	Low Risk

RESULTS



The pictures indicate some of the actual and predicted seismic-induced landslides that have occurred in the last few years in the northern part of Ok Tedi mining area. Most of these failures have been shallow landslides (0 – 2.5 m) as shown in panels 1, 2, 3, 5 and 6. Panels 4 and 7 show relatively deeper landslides (3-5 m thick) in the brecciated / sheared mudstone (panel 4) and limestone (panel 7). The shallow landslide in panel 2 occurred in 2017 and involved a slip of less than 1 m thick weathered and oxidized regolith during a small earthquake of approximately magnitude 5. The failure was enhanced by drilling activities, which involved uncontrolled disposal of drilling fluids. The discussed 2018 earthquake (M 7.5) is known to have induced significant slope failure in the area. The Wellington exploration camp is generally located on a low-risk zone (FoS ~ 3). Although rainfall (associated with erosion and water seeping/toe undercutting of slopes) is also known to be solely responsible for a significant number of slope failures in the area, most of the slope failures are a combination of both resulting in the existence of many active shallow landslides in the area.

CLOSING

Next Step: Action

- Infrastructure planning
- Settlement/homes relocation
- Community safety awareness

Examples and Similar Projects Elsewhere

A list of similar projects can be viewed online simply by searching "**seismic microzonation and seismic slope stability analysis in gis**" on Google

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